
Power Supplies

A Solving Manual

IESE Manual 3 of 3

Typeset in L^AT_EX with circuitikz and pgfplots.

How to use this manual

A power supply turns an available energy source (mains, battery, solar, dynamo) into the stable voltage a circuit needs. This manual is organised by the four building blocks the course studies — the classical transformer supply, the power-factor problem, the linear regulator, and the switching regulator — and for each *type* of problem it gives a **Protocol** (a numbered recipe), one fully **Worked Example** (the canonical one), and **Exercises** with solutions. Theory sits at the point of use in one paragraph; the full derivation is in a cross-referenced **appendix**.

Key Idea

Two numbers run through everything here. *Ripple* ΔV (how far the output sags between recharges) and *efficiency* $\eta = P_{out}/P_{in}$ (how much input power reaches the load). Linear regulators give clean output but poor η ; switching regulators give high η at the cost of high-frequency ripple. Every design trades these off.

Contents

How to use this manual	ii
1 Specifications: reading the data sheet	1
2 The classical (transformer) power supply	2
2.1 Transformer and rectifier	2
2.2 Sizing the filter capacitor	3
3 Power factor and its correction	5
4 Linear voltage regulators	8
4.1 Shunt (parallel) and series topologies	8
4.2 The detailed series regulator	8
5 Switching regulators	11
5.1 Transfer functions from volt-second balance	11
6 Laboratory: Buck converter	14
6.1 Static measurements and the CCM/DCM test	14
6.2 Load regulation	14
6.3 Dynamic waveform and switching parameters	15
6.4 Efficiency	15
A Ripple and conduction time	16
B Power-factor derivation	17
C Converter transfer functions (CCM)	18
D Reference data	19
E Example coverage map	20

Chapter 1

Specifications: reading the data sheet

Before solving anything you must read the spec correctly. The recurring quantities are *regulation* (how much the output moves) and its companion *sensitivity*, plus *ripple rejection* and *efficiency*.

Protocol 1.1 Extract regulation, sensitivity, ripple rejection

1. **Load regulation:** the output change ΔV_{out} for a stated load-current change ΔI_{out} . *Load sensitivity* $S_I = \Delta V_{out}/\Delta I_{out}$ — an output resistance, in Ω .
2. **Line regulation:** the output change ΔV_{out} for an input change ΔV_{in} . *Line sensitivity* $S_V = \Delta V_{out}/\Delta V_{in}$.
3. **Ripple rejection** in dB: $20 \log_{10}(\Delta V_{in}/\Delta V_{out})$.
4. **Efficiency** $\eta = P_{out}/P_{in}$; dissipated power $P_d = P_{in} - P_{out}$ becomes heat (size the heat sink from it).

Worked Example 1.1 Reading a regulator's numbers

A regulator shows $\Delta V_{out} = 7 \text{ mV}$ for $\Delta V_{in} = 7.5 \text{ V}$ to 25 V (i.e. $\Delta V_{in} = 17.5 \text{ V}$): ripple rejection $= 20 \log_{10}(17.5/0.007) = 20 \log_{10}(2500) = 68 \text{ dB}$. For a load step $\Delta I_{out} \approx 1.5 \text{ A}$ giving $\Delta V_{out} = 30 \text{ mV}$, the output resistance is $S_I = 0.03/1.5 = 20 \text{ m}\Omega$.

Exercise 1.1 A datasheet line by line

A 5 V regulator's table reads: $\Delta V_{out} = 10 \text{ mV}$ for I_{out} from 5 mA to 1.5 A; $\Delta V_{out} = 3 \text{ mV}$ for V_{in} from 8 V to 12 V. (a) Load and line sensitivity, and the ripple rejection. (b) At $V_{in} = 12 \text{ V}$, $I_{out} = 1 \text{ A}$: efficiency and dissipated power.

Solution

(a) $S_I = 0.010/1.495 \approx 6.7 \text{ m}\Omega$; $S_V = 0.003/4 = 0.75 \text{ mV/V}$; ripple rejection $= 20 \log_{10}(4/0.003) = 62.5 \text{ dB}$. (b) $P_{out} = 5 \times 1 = 5 \text{ W}$, $P_{in} \approx 12 \times 1 = 12 \text{ W}$: $\eta = 5/12 = 41.7 \%$ and $P_d = 7 \text{ W}$ of heat — the heat-sink question of Chapter 4.

Watch Out

Static (DC) regulation figures are not the whole story: during a *transient* load step the output can deviate by hundreds of mV — far more than the 25 mV to 30 mV static value. Always check the transient response, not just the steady-state table.

Chapter 2

The classical (transformer) power supply

The classical supply is a chain of blocks: mains, protection, EMC filter, transformer (step-down + isolation), rectifier (AC→unipolar), capacitor filter (unipolar→pseudo-DC), and a regulator. It gives a very clean output but poor efficiency and a pulsed input current.

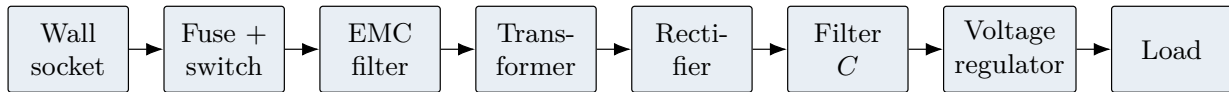


Figure 2.1: Block diagram of the classical power supply. 230 V_{RMS}@50 Hz → a clean low-voltage DC for the load.

2.1 Transformer and rectifier

The transformer scales voltage by its turns ratio and isolates galvanically: $V_S/V_P = N_S/N_P$ and (ideally lossless) $V_P I_P = V_S I_S$. A *full-wave* bridge rectifier inverts the negative half-cycles, so its output ripples at *twice* the line frequency (period $T/2$); a half-wave rectifier passes only the positive halves (period T). Diodes drop 1 V to 2 V.

The design runs *backwards*, from the load to the wall: the regulator needs a minimum input (its output plus dropout), the ripple eats into that margin, the bridge eats two diode drops off the secondary peak, and the mains itself may sag 10 % — each stage adds its demand to the one before.

Protocol 2.1 Size the transformer and rectifier

1. **Peak at the filter:** pick V_{MAX} so the ripple valley clears the regulator, $V_{MAX} - \Delta V \geq V_{out} + V_{DO,min}$ (plus margin).
2. **Secondary voltage:** the bridge drops two diodes, so the secondary peak must be $V_{MAX} + 2V_D$, i.e. $V_{S,RMS} = (V_{MAX} + 2V_D)/\sqrt{2}$.
3. **Worst-case mains:** the result must hold at mains -10% , so divide by 0.9, then pick the next standard secondary; turns ratio $N_P/N_S = V_{P,RMS}/V_{S,RMS}$.
4. **Rate the diodes:** reverse voltage $\geq V_{MAX}$ (with margin) and repetitive peak current $\geq I_{peak}$ from the filter protocol below.

Worked Example 2.1 Secondary for a 5 V/100 mA supply

Target: a 7805 ($V_{DO,min} = 2$ V) fed through a bridge ($V_D = 1$ V) with ripple $\Delta V = 0.9$ V (next section). *Step 1:* the valley must stay above $5 + 2 = 7$ V; choosing $V_{MAX} = 9$ V leaves a valley of $9 - 0.9 = 8.1$ V — 1.1 V of margin. *Step 2:* secondary peak = $9 + 2 = 11$ V, so $V_{S,RMS} = 11/\sqrt{2} = 7.8$ V. *Step 3:* at mains -10% that becomes $7.8/0.9 = 8.6$ V — pick a standard 9 V_{RMS} secondary, ratio $N_P/N_S = 230/9 \approx 26$. *Check* at nominal mains: peak

$$= 9\sqrt{2} - 2 = 10.7 \text{ V, valley } 9.8 \text{ V} \text{ — fine (the regulator absorbs the extra).}$$

Exercise 2.1

Size the secondary for 12 V/0.5 A from a 7812 ($V_{DO,min} = 2 \text{ V}$), bridge with $V_D = 1 \text{ V}$, ripple $\Delta V \leq 1 \text{ V}$, mains $230 \pm 10\%/50 \text{ Hz}$. (a) The filter capacitor. (b) The secondary RMS voltage and ratio. (c) Why would a *half-wave* rectifier double the capacitor?

Solution

(a) Full-wave, $T_{dis} = T/2 = 10 \text{ ms}$: $C = I_{out}T_{dis}/\Delta V = 0.5 \times 0.01/1 = 5 \text{ mF}$. (b) Valley $\geq 12 + 2 = 14 \text{ V} \Rightarrow V_{MAX} = 15 \text{ V}$; secondary peak $= 15 + 2 = 17 \text{ V}$, $V_{S,RMS} = 17/\sqrt{2} = 12.0 \text{ V}$; at -10% : $12.0/0.9 = 13.4 \text{ V}$ — pick 15 V_{RMS} (standard), ratio $\approx 230/15 = 15.3$. (c) Half-wave discharges for the whole period T instead of $T/2$: same ΔV needs twice the charge, hence $C = 10 \text{ mF}$.

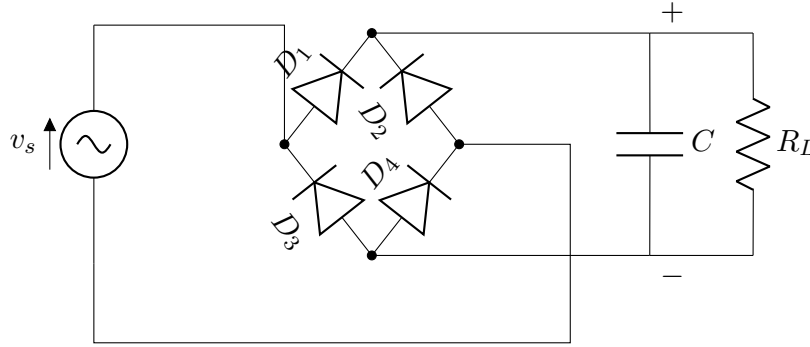


Figure 2.2: Full-wave bridge rectifier with capacitor filter and load. The capacitor holds the peak and discharges into the load between peaks, leaving a residual ripple ΔV .

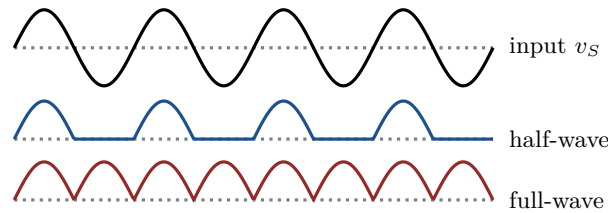


Figure 2.3: Rectification. The half-wave output keeps only the positive half-cycles (ripple period T); the full-wave output flips the negative halves up, so it ripples at *twice* the line frequency (period $T/2$). Real diodes also shave 1 V to 2 V off each peak.

2.2 Sizing the filter capacitor

The filter capacitor charges briefly near each peak and discharges (approximately linearly) into the load for the rest of the period. The ripple is $\Delta V = I_{out}T_{dis}/C$ with $T_{dis} = T/2$ (full-wave) or T (half-wave). Charge balance fixes the peak input current. Derivation: Appendix A.

Protocol 2.2 Filter ripple, conduction time and peak current

1. **Ripple:** $\Delta V = \frac{I_{out}}{C} \cdot T_{dis}$, $T_{dis} = T/2$ (full-wave) or T (half-wave).

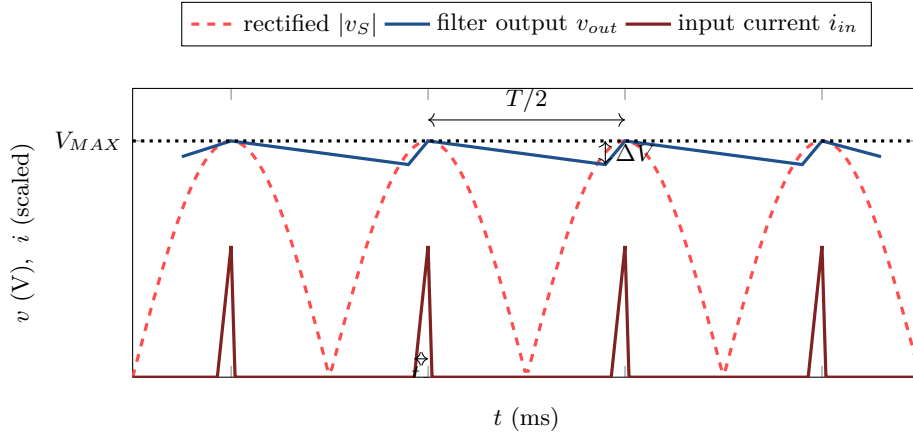


Figure 2.4: Filter action (full-wave). The capacitor charges in a brief pulse near each peak V_{MAX} , then discharges almost linearly into the load for the rest of the half-period, leaving the residual ripple ΔV . The recharge is crammed into the short conduction time t_c , so the input current i_{in} arrives as tall narrow spikes of height I_{peak} . For the worked example below ($V_{MAX} = 9\text{ V}$, $I_{out} = 100\text{ mA}$, $C = 2.2\text{ mF}$): $\Delta V = 0.9\text{ V}$, $t_c = 1.44\text{ ms}$, $I_{peak} = 2.77\text{ A}$. A larger C flattens ΔV but shortens t_c , pushing I_{peak} up.

2. **Conduction time:** from where the rising sine re-reaches $V_{MAX} - \Delta V$, $t_c = \frac{T}{2\pi} \cos^{-1} \frac{V_{MAX} - \Delta V}{V_{MAX}}$.
3. **Peak charging current:** charge in = charge out over a period, $\frac{1}{2}t_c I_{peak} = I_{out}T \Rightarrow I_{peak} = \frac{2I_{out}T}{t_c}$.
4. **Trade-off:** larger $C \Rightarrow$ smaller ΔV but *larger* I_{peak} (shorter t_c). Choose against both specs.

Worked Example 2.2 Full-wave filter, $C = 2.2\text{ mF}$

$f = 50\text{ Hz}$ ($T = 20\text{ ms}$), $V_{MAX} = 9\text{ V}$, $I_{out} = 100\text{ mA}$. Using $T_{dis} = T$, the ripple is $\Delta V = \frac{I_{out}T}{C} = \frac{0.1 \cdot 0.02}{2.2 \times 10^{-3}} = 0.9\text{ V}$. Conduction time $t_c = \frac{0.02}{2\pi} \cos^{-1} \frac{9 - 0.9}{9} = 1.44\text{ ms}$; peak current $I_{peak} = \frac{2 \cdot 0.1 \cdot 0.02}{1.44 \times 10^{-3}} = 2.77\text{ A}$.

Exercise 2.2

Repeat with $C' = 4.7\text{ mF}$ (same f, V_{MAX}, I_{out}). What happens to ΔV and to I_{peak} ?

Solution

$\Delta V' = \frac{0.1 \cdot 0.02}{4.7 \times 10^{-3}} = 0.43\text{ V}$ (smaller); $t'_c = \frac{0.02}{2\pi} \cos^{-1} \frac{9 - 0.43}{9} = 0.98\text{ ms}$; $I'_{peak} = \frac{2 \cdot 0.1 \cdot 0.02}{0.98 \times 10^{-3}} = 4.1\text{ A}$ (larger). Lower ripple, higher current peak — the trade-off in action.

Chapter 3

Power factor and its correction

The rectifier-capacitor front end draws current in short pulses, far from the sinusoidal in-phase current an ideal resistive load would draw. Two effects lower the **power factor** $PF = P/S$: a *phase shift* between voltage and current (displacement, $\cos \varphi$) and *harmonic distortion* ($1/\sqrt{1 + THD^2}$). Regulations require $PF \gtrsim 0.9$.

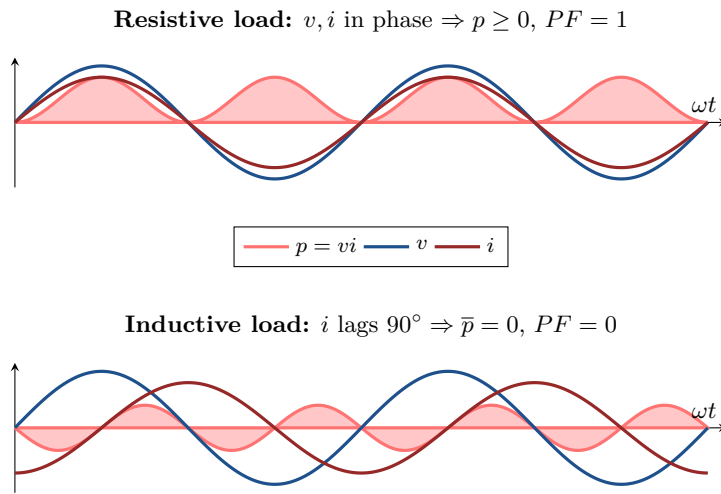


Figure 3.1: Instantaneous power $p = vi$ (shaded). With a resistive load v and i are in phase, so p never goes negative — real power flows continuously ($PF = 1$). With a purely inductive load i lags by 90° , so p is positive and negative in equal measure: the average power is zero ($PF = 0$) yet the current still loads the line. A real R – L load lies between, at $PF = \cos \varphi$.

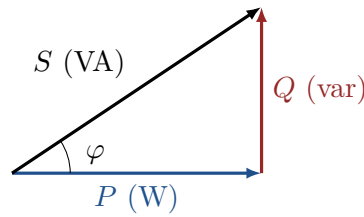


Figure 3.2: Power triangle: $S = \sqrt{P^2 + Q^2}$, $PF = \cos \varphi = P/S$.

Protocol 3.1 Compute the power factor; correct it

1. **Current:** for a series R – L load, $I_S = \frac{V_S}{R + j\omega L}$; magnitude $|I_S| = \frac{V_S}{\sqrt{R^2 + (\omega L)^2}}$, phase $\angle I_S = -\arctan \frac{\omega L}{R}$.
2. **Powers:** $S = V_{RMS} I_{RMS}$; $P = V_{RMS} I_{RMS} \cos \varphi$; $PF = P/S = \cos \varphi$.
3. **Correct (PFC):** add a *parallel* capacitor to cancel the reactive part. For the

R – L load, $\angle I_S = 0$ requires $C = \frac{L}{\omega^2 L^2 + R^2}$ (Appendix B). The load voltage and current are unchanged; only the source current is brought into phase.

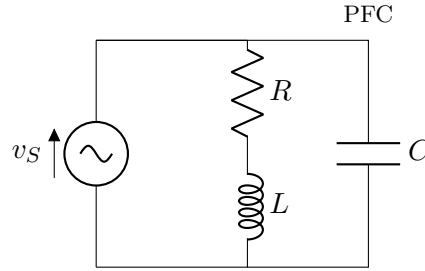


Figure 3.3: Power-factor correction: a capacitor in parallel with the inductive load cancels the phase shift seen by the source.

Note

Where $C = L/(R^2 + \omega^2 L^2)$ comes from. Work in admittances. The R – L load has

$$Y_{load} = \frac{1}{R + j\omega L} = \frac{R - j\omega L}{R^2 + \omega^2 L^2},$$

whose imaginary part (susceptance) is the lagging $-\frac{\omega L}{R^2 + \omega^2 L^2}$. A parallel capacitor adds $+\omega C$. Unity power factor means the *total* susceptance vanishes, $\omega C = \frac{\omega L}{R^2 + \omega^2 L^2}$, hence $C = \frac{L}{R^2 + \omega^2 L^2}$. The load's *real* current — and therefore the power it consumes — is untouched; only the reactive current is moved off the grid and onto the local LC pair. (With $R \rightarrow 0$ this collapses to the resonance condition $\omega^2 LC = 1$, the pure-inductor case below.)

Worked Example 3.1 R – L load and its correction

$R = 3\ \Omega$, $L = 10.6\text{ mH}$, $V_S = 480\text{ V}_{\text{RMS}}$ at $f = 60\text{ Hz}$ ($\omega = 377\text{ rad/s}$). Then $\omega L \approx 4$, $\sqrt{R^2 + (\omega L)^2} = \sqrt{9 + 16} = 5\ \Omega$, so $I_S = 480/5 = 96\text{ A}$, $\varphi = \arctan(4/3) = 53.1^\circ$. $S = 480 \cdot 96 = 46.08\text{ kVA}$, $P = S \cos \varphi = 27.67\text{ kW}$, $PF = 0.6$. **Correction:** $C = \frac{L}{\omega^2 L^2 + R^2} = \frac{0.0106}{25} = 424.5\ \mu\text{F}$; the source current drops to $I_S = 58\text{ A}$ and $PF = 1$ — same load power, far less current drawn from the grid.

Exercise 3.1

A purely inductive load $L = 10.6\text{ mH}$ (no R), same source. Find I_S and PF , and explain the energy flow.

Solution

$|I_S| = V_S/(\omega L) = 480/4 = 120\text{ A}$, $\varphi = 90^\circ$, $PF = \cos 90^\circ = 0$. The active power is zero: energy sloshes into the inductor each quarter-cycle and back out the next, dissipating only in the grid wiring — the load uses no net power yet still loads the network.

Exercise 3.2 Compensate the pure inductor

Correct the previous exercise's load ($L = 10.6 \text{ mH}$, no R , $V_S = 480 \text{ V}_{\text{RMS}}$, $\omega = 377 \text{ rad/s}$): find the parallel C that brings the source current to zero, and explain where the inductor's energy then comes from.

Solution

With $R = 0$ the PFC formula reduces to resonance: $C = \frac{L}{\omega^2 L^2} = \frac{1}{\omega^2 L} = \frac{1}{377^2 \times 0.0106} = 663.8 \text{ }\mu\text{F}$, and ideally $I_S = 0$. The LC pair is a parallel resonator at 60 Hz : the energy simply sloshes *between* the capacitor and the inductor each quarter-cycle, and the grid no longer carries that exchange. (Sanity check: $R \rightarrow 0$ in $C = L/(R^2 + \omega^2 L^2)$ gives the same result.)

Chapter 4

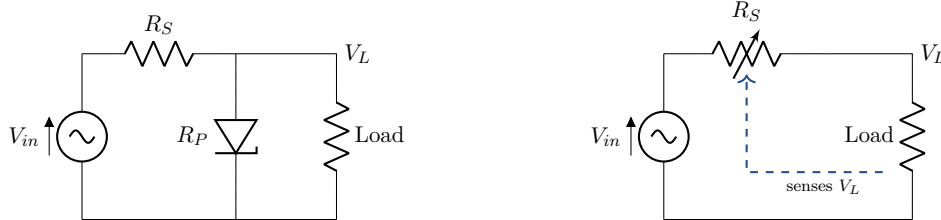
Linear voltage regulators

A linear regulator holds its output constant by behaving as a *variable voltage divider* between the input and ground: a series element R_S and a shunt element R_P , one of which is continuously trimmed to absorb line (ΔV_{in}) and load (ΔI_{out}) variations so that V_L stays put. Which element does the trimming gives the two classic topologies.

4.1 Shunt (parallel) and series topologies

Parallel (shunt) regulator (Fig. 4.1a). The series resistor R_S is *fixed*; the variable element R_P sits *across* the load and is realised by a **Zener diode**, which pins its own terminal voltage. When the load draws less, the shunt swallows the difference, so the current from the source $I \approx (V_{in} - V_L)/R_S$ stays essentially *constant* whatever the load does. It is trivially simple and short-circuit-friendly, but it wastes that full current at all times — fit only for small or fixed loads. (That Zener shunt is exactly the voltage *reference* the series type uses internally.)

Series regulator (Fig. 4.1b). Now the *series* element R_S is the variable one and no shunt is needed: an amplifier senses V_L and trims R_S (a pass transistor) through **negative feedback**, so only the load current flows. It is far more efficient, and is the standard choice the rest of this chapter develops.



(a) Parallel (shunt): fixed R_S , variable R_P (a Zener) (b) Series: variable R_S , set by feedback from V_L

Figure 4.1: The two linear-regulator topologies. **(a)** A *shunt* regulator fixes the series resistor and varies a parallel element (a Zener) across the load, so it draws roughly constant current regardless of load. **(b)** A *series* regulator varies the series element under feedback from V_L so only the load current flows (detailed in Fig. 4.2).

4.2 The detailed series regulator

In the practical series regulator the variable resistance is the controlled on-resistance of an *npn* pass transistor, and the amplifier is a real op-amp that compares a divided copy of V_L against a reference (a Zener or band-gap). At the op-amp's virtual short the inverting input sits at V_{REF} , while the divider makes it $V_{REF} = V_L R_2/(R_1 + R_2)$; equating the two gives

$$V_L = V_{REF} \left(1 + \frac{R_1}{R_2} \right),$$

independent of V_{in} and I_{out} to the extent the loop gain is high. The regulator's *ripple rejection* is simply that loop gain at the ripple frequency — the PSRR number on the data sheet (Chapter 1).

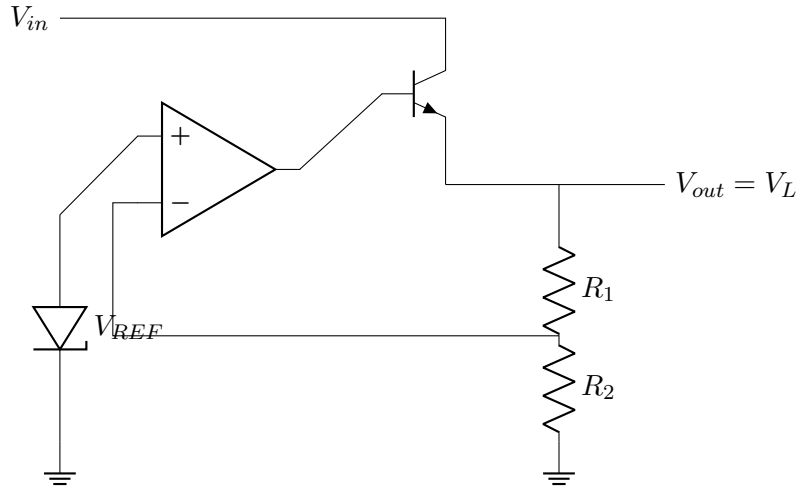


Figure 4.2: Series linear regulator: the op-amp drives the pass transistor so that the divided output equals V_{REF} (a Zener/band-gap), giving $V_L = V_{REF}(1 + R_1/R_2)$.

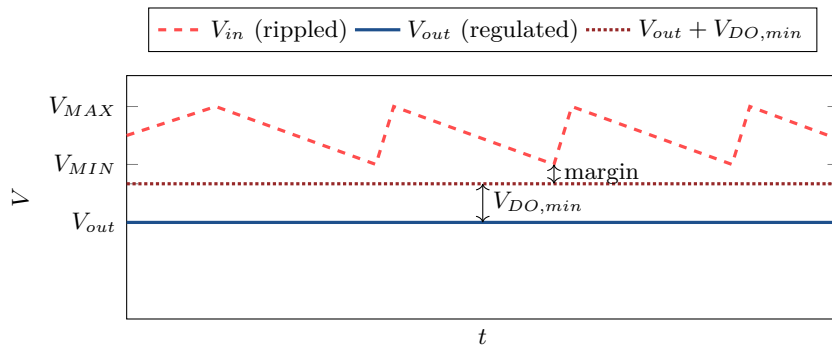


Figure 4.3: Dropout headroom. The regulator input is the rippled filter output, swinging between V_{MAX} and V_{MIN} . Regulation holds only while $V_{in} > V_{out} + V_{DO,min}$ at *every* instant — so the ripple valley V_{MIN} must stay above the dropout floor. Here $V_{MIN} = 8\text{ V}$ clears the floor $5 + 2 = 7\text{ V}$ by 1 V ; shrink the filter C until $V_{MIN} < 7\text{ V}$ and the output would sag with the input each cycle.

Protocol 4.1 Linear-regulator output, dissipation and efficiency

1. **Output:** $V_L = V_{REF}(1 + R_1/R_2)$ (set by the divider).
2. **Dropout:** $V_{out} = V_{in} - V_{DO}$; require $V_{in} > V_{out} + V_{DO,min}$ of the rippled input, and $V_{DO} \geq V_{DO,min}$ ($\approx 2\text{ V}$ for a 78xx).
3. **Dissipation:** $P_{reg} \approx (V_{in} - V_{out})I_{out} = V_{DO}I_{out}$ — all heat.
4. **Efficiency:** $\eta = V_{out}/V_{in} = 1 - V_{DO}/V_{in}$; $\eta_{MAX} = V_{out}/(V_{out} + V_{DO,min})$.

Worked Example 4.1 Efficiency of a 78xx ($V_{DO,min} = 2\text{ V}$)

$$\eta_{MAX} = V_{out}/(V_{out} + 2):$$

V_{out} (V)	1.5	3.3	5	9	12
η_{MAX}	42.9%	62.3%	71.4%	81.8%	85.7%

Low output voltages waste most of the input power as heat — the central reason switching regulators exist.

Worked Example 4.2 Dissipation: does it need a heat sink?

A 7805 supplies $I_{out} = 1$ A from $V_{in} = 12$ V: $P_{reg} = (V_{in} - V_{out})I_{out} = (12 - 5) \times 1 = 7$ W and $\eta = 5/12 = 42\%$. A bare TO-220 has a junction-to-ambient thermal resistance $R_{th} \approx 60$ K/W, so the junction would sit $7 \times 60 = 420$ K above ambient — far past the 125°C limit. With a 10 K/W heat sink the rise is 70 K: $T_j \approx 25 + 70 = 95^\circ\text{C}$ — acceptable. The heat sink is not optional here.

Exercise 4.1

Same 7805. (a) At $V_{in} = 8$ V, $I_{out} = 200$ mA: dissipation, efficiency, and is the bare package (60 K/W) enough at $T_{amb} = 25^\circ\text{C}$? (b) What is the lowest rippled-input *valley* that still regulates?

Solution

(a) $P_{reg} = (8 - 5) \times 0.2 = 0.6$ W, $\eta = 5/8 = 62.5\%$; junction rise $0.6 \times 60 = 36$ K $\Rightarrow T_j \approx 61^\circ\text{C}$ — fine without a heat sink. Reducing the input headroom helps twice: less heat *and* better efficiency. (b) $V_{MIN} = V_{out} + V_{DO,min} = 5 + 2 = 7$ V — below it the output sags with the ripple (Fig. 4.3).

Note

A **Low-Drop-Out (LDO)** regulator replaces the common-collector NPN with a PNP or pMOS pass device, so V_{DO} can be only a few mV — recovering efficiency at low V_{out} , at the cost of trickier stability (an external output capacitor sets a compensation pole).

Chapter 5

Switching regulators

A switching regulator transfers energy with *only* reactive components (inductor, capacitors) and switches, so ideally it dissipates nothing ($\eta \rightarrow 100\%$, in practice 80 % to 90 %). A primary switch SW_1 (an nMOS) chops at frequency f_{SW} with duty cycle $\delta = T_{ON}/T_{SW}$; a secondary switch SW_2 (a diode) conducts in the opposite phase. The inductor current ramps up and down, triangular in shape: in **CCM** it never reaches zero, in **DCM** it does.

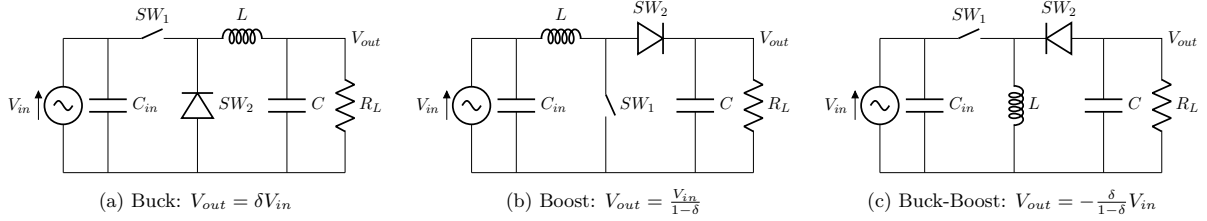


Figure 5.1: The three basic DC–DC converters (SW_1 = nMOS, SW_2 = diode). The Switching Controller (not shown) sets δ by PWM.

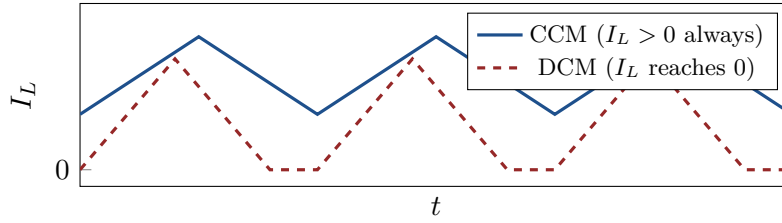


Figure 5.2: Inductor current: triangular, ramping up while SW_1 is closed and down while open. DCM appears at low output current.

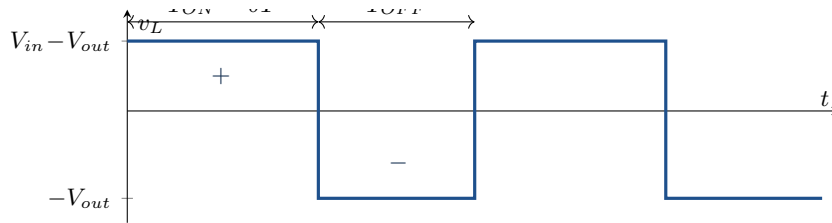


Figure 5.3: Inductor voltage in a Buck (CCM). With SW_1 closed ($T_{ON} = \delta T$) the inductor sees $+(V_{in} - V_{out})$; with SW_1 open the diode SW_2 conducts and it sees $-V_{out}$. Each constant level makes i_L ramp *linearly* (Fig. 5.2), and steady state forces the volt–seconds to balance, $(V_{in} - V_{out})\delta = V_{out}(1 - \delta)$, i.e. $V_{out} = \delta V_{in}$.

5.1 Transfer functions from volt–second balance

In steady state the inductor current returns to the same value every cycle, so its net change over a period is zero — equivalently the *average* inductor voltage is zero. This is **volt–second balance**:

the area v_L spends positive must equal the area it spends negative, $\langle v_L \rangle = V_{ON} \delta T + V_{OFF}(1-\delta)T = 0$. Applying it to each topology gives the transfer function directly — no calculus, just equal areas.

Buck On: $v_L = V_{in} - V_{out}$ for δT ; off: $v_L = -V_{out}$ for $(1-\delta)T$. Balance $(V_{in} - V_{out})\delta = V_{out}(1-\delta)$ gives

$$V_{out} = \delta V_{in} \quad (\text{step-down, } 0 < \delta < 1).$$

Boost On, SW_1 ties the inductor across the input: $v_L = V_{in}$ for δT . Off, the inductor stacks on V_{in} and pushes through SW_2 : $v_L = V_{in} - V_{out}$ for $(1-\delta)T$. Balance $V_{in}\delta = (V_{out} - V_{in})(1-\delta)$ gives

$$V_{out} = \frac{V_{in}}{1-\delta} \quad (\text{step-up}).$$

Buck-Boost

On: $v_L = V_{in}$ for δT . Off, the inductor discharges into the output through SW_2 : $v_L = V_{out}$ for $(1-\delta)T$ (with $V_{out} < 0$). Balance $V_{in}\delta = -V_{out}(1-\delta)$ gives

$$V_{out} = -\frac{\delta}{1-\delta} V_{in} \quad (\text{step up or down, inverted}).$$

The same equal-areas idea fixes the inductor-current ripple: during the ON time the constant voltage V_{ON} ramps i_L by $\Delta I_L = V_{ON} \delta T / L$, which is the quantity the CCM check below compares against the load.

Protocol 5.1 Choose the topology and solve the duty cycle

1. **Pick topology** from V_{out} vs V_{in} : step-down $\rightarrow$ Buck; step-up $\rightarrow$ Boost; invert/either $\rightarrow$ Buck-Boost.
2. **Solve δ** from the CCM transfer function (Fig. 5.1); get the *range* of δ from the V_{in} range.
3. **Check CCM**: find the minimum L that keeps $I_L > 0$ at the lowest load (Appendix C); below it the converter enters DCM and load regulation degrades.
4. **Output ripple**: $\Delta V_{out} = \Delta I_C \cdot \text{ESR}$ of the output capacitor.

Worked Example 5.1 $V_{in} = 9\text{ V} \pm 1\text{ V}$, three targets

Buck $V_{out} = 5\text{ V}$: $\delta = V_{out}/V_{in} = 5/9 = 55.5\%$; range $\delta_{MAX} = 5/8 = 62.5\%$, $\delta_{MIN} = 5/10 = 50\%$. **Boost** $V_{out} = 15\text{ V}$: $\delta = 1 - V_{in}/V_{out} = 1 - 9/15 = 40\%$; range $46.7\% \dots 33.3\%$. **Buck-Boost** $V_{out} = -9\text{ V}$: $\delta = \frac{V_{out}}{V_{out} - V_{in}} = \frac{-9}{-18} = 50\%$; range $52.9\% \dots 47.4\%$. The controller adjusts δ as V_{in} drifts.

Worked Example 5.2 Minimum L for CCM (Buck)

The Buck question: find the smallest L that keeps the converter in CCM at the lightest load. The inductor ripple is $\Delta I_L = \frac{(V_{in} - V_{out})\delta}{L f_{SW}}$, and CCM requires $\Delta I_L \leq 2I_{out}$ (Appendix C), so $L_{min} = \frac{(V_{in} - V_{out})\delta}{2I_{out,min}f_{SW}}$. For the running Buck ($V_{in} = 9\text{ V}$, $V_{out} = 5\text{ V}$, $\delta = 0.556$) with $f_{SW} = 100\text{ kHz}$ and $I_{out,min} = 50\text{ mA}$: $L_{min} = \frac{4 \times 0.556}{2 \times 0.05 \times 10^5} = 222\text{ }\mu\text{H}$; pick the next standard $330\text{ }\mu\text{H}$ ($\Delta I_L = 67\text{ mA} < 100\text{ mA}$ — CCM holds). Below $I_{out,min}$ the converter

enters DCM and V_{out} rises unless the controller trims δ .

Exercise 5.1

A 12 V rail must come from a battery $V_{in} = (5 \pm 0.5) \text{ V}$, $f_{SW} = 200 \text{ kHz}$. (a) Topology and the δ range. (b) With a capacitor-current swing $\Delta I_C = 1.2 \text{ A}$ and $\text{ESR} = 50 \text{ m}\Omega$, the output ripple. (c) Why is this topology's feedback loop deliberately made *slow*?

Solution

(a) $V_{out} > V_{in}$: **Boost**, $\delta = 1 - V_{in}/V_{out}$; over $V_{in} = 4.5 \dots 5.5$: $\delta = 62.5\% \dots 54.2\%$ (nominal 58.3%). (b) $\Delta V_{out} = \Delta I_C \cdot \text{ESR} = 1.2 \times 0.05 = 60 \text{ mV}$ — in a switching supply the ESR, not the capacitance, usually sets the ripple. (c) The Boost is prone to instability, so its loop gain is designed with a low bandwidth: it reacts slowly to load changes but does not oscillate.

Note

The **Flyback** converter is a Buck-Boost whose inductor becomes a transformer: it adds galvanic isolation, a turns-ratio degree of freedom, a selectable output polarity, and multiple outputs. A **Boost** run from the rectified mains is the standard active **PFC** stage: by shaping the inductor current to follow the input voltage it presents a near-resistive load to the grid.

Chapter 6

Laboratory: Buck converter

This lab characterises a real DC–DC **buck** converter (an XL4015 module), putting Chapter 5 to the test: measure input/output voltages under load and no load, decide whether it runs in **CCM** or **DCM**, read the switching waveform, and map the efficiency. Instruments: an HP 34401A multimeter, a Rigol DP832 supply, a Rigol DS1054 DSO, and an adjustable high-power resistive load.

Key Idea

A switcher's behaviour splits in two: *static* (DC in/out, regulation, efficiency — a multimeter job) and *dynamic* (the switching ripple — a scope job). The single most important question is which conduction mode it is in, because every dynamic formula depends on the answer.

6.1 Static measurements and the CCM/DCM test

Read V_{in} and V_{out} with and without the load; with a load R_L the output current is $I_{out} = V_{out}/R_L$. The boundary between continuous (CCM) and discontinuous (DCM) conduction is the *critical* current — half the inductor-current ripple:

$$I_B = \frac{V_{out} (V_{in} - V_{out})}{2 L f_{sw} V_{in}}.$$

Protocol 6.1 Decide CCM vs DCM

1. Measure V_{in} , V_{out} and (loaded) compute $I_{out} = V_{out}/R_L$.
2. Compute the boundary current I_B from the nominal L and f_{sw} .
3. If $I_{out} > I_B$ the converter is in **CCM**; if $I_{out} < I_B$ it is in **DCM**. No load ($I_{out} \approx 0$) is always DCM.

Worked Example 6.1 Two operating points

Case 1 ($V_{in} = 8\text{ V}$, $V_{out} = 5\text{ V}$, $R_L \approx 30.5\ \Omega$): loaded $I_{out} \approx 0.16\text{ A}$ exceeds the boundary $I_B \approx 0.11\text{ A}$, so it runs in **CCM**. **Case 2** ($V_{in} = 24\text{ V}$, $V_{out} = 5\text{ V}$, $R_L \approx 50\ \Omega$): $I_{out} \approx 0.10\text{ A}$ is below $I_B \approx 0.23\text{ A}$, so despite the heavier line it stays in **DCM** — a light load at a high step-down ratio keeps you discontinuous.

6.2 Load regulation

Load regulation is the output droop from no-load to full-load, $\Delta V_{out} = V_{out, \text{no load}} - V_{out, \text{load}}$. Case 1 gave $\Delta V_{out} \approx 68\text{ mV}$ on a 5 V output ($\sim 1.4\%$) — read it against the spec (Chapter 1).

6.3 Dynamic waveform and switching parameters

In CCM the steady-state duty cycle is $\delta = V_{out}/V_{in}$, and from it

$$T_{sw} = \frac{1}{f_{sw}}, \quad T_{on} = \delta T_{sw}, \quad T_{off} = (1 - \delta)T_{sw}, \quad \Delta I_L = \frac{V_{in} - V_{out}}{L} T_{on}, \quad V_{ripple} = \Delta I_L \cdot \text{ESR}.$$

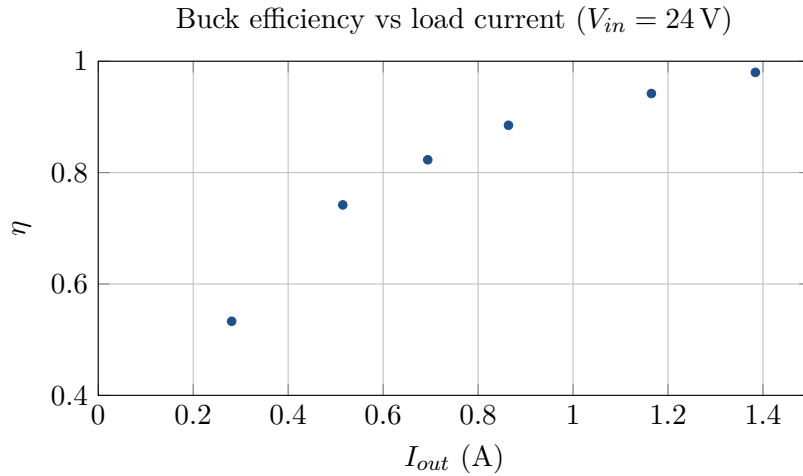
Read the ripple on the DSO in **AC coupling** (so the small ripple is not lost on top of the DC level), and recover $f_{sw} = 1/T_{sw}$ and $\delta = T_{on}/T_{sw}$ from the trace.

Watch Out

The XL4015's switching frequency is set by an *internal* clock you cannot probe directly. Its nominal 180 kHz actually ranges 144 kHz to 216 kHz; the measured $f_{sw} \approx 207$ kHz is in-spec. So the time-domain quantities (T_{sw}, T_{on}, δ) computed from the *nominal* frequency will not match the measured ones — the voltage quantities (V_{out}, V_{ripple}) still do.

6.4 Efficiency

Efficiency is $\eta = P_{out}/P_{in} = V_{out}I_{out}/(V_{in}I_{in})$, with I_{in} read back from the supply. Swept against output current (here at $V_{in} = 24$ V, varying the load) it rises steeply out of light load and flattens high — the shape on the XL4015 datasheet.



Note

Light-load efficiency is poor (Case 2 reached only $\eta \approx 0.34$): the converter's own quiescent and switching losses are a large fraction of a small output power. This is exactly why DCM/light-load operation is penalised, and why the curve climbs toward ~ 0.98 as the load grows.

Appendix A

Ripple and conduction time

Modelling the discharge as linear at constant load current I_{out} , the charge removed in one discharge interval T_{dis} is $Q = I_{out}T_{dis} = C\Delta V$, hence $\Delta V = I_{out}T_{dis}/C$ (with $T_{dis} = T/2$ full-wave, T half-wave). The capacitor recharges only while the rectified sine exceeds the sagged output, i.e. from the angle where $V_{MAX} \cos \theta = V_{MAX} - \Delta V$, giving $t_c = \frac{T}{2\pi} \cos^{-1} \frac{V_{MAX} - \Delta V}{V_{MAX}}$. Equating the triangular charge pulse area $\frac{1}{2}t_c I_{peak}$ to the charge delivered to the load over a period $I_{out}T$ gives $I_{peak} = 2I_{out}T/t_c$.

Appendix B

Power-factor derivation

Real power. With $v = V_S \sin \omega t$ and $i = I_S \sin(\omega t - \varphi)$, $P = \frac{1}{T} \int_0^T vi \, dt$. Using $\sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$ the double-frequency term integrates to zero, leaving $P = \frac{1}{2} V_S I_S \cos \varphi = V_{RMS} I_{RMS} \cos \varphi$. **PFC capacitor.** Adding C in parallel, $I_S = V_S(j\omega C + \frac{1}{R + j\omega L})$; setting $\Im\{I_S\} = 0$ gives $\omega C = \frac{\omega L}{R^2 + (\omega L)^2}$, i.e. $C = \frac{L}{R^2 + \omega^2 L^2}$.

Appendix C

Converter transfer functions (CCM)

In steady state the inductor's volt-second balance (average $V_L = 0$ over a period) gives the CCM ratios: Buck $V_{out} = \delta V_{in}$; Boost $V_{out} = V_{in}/(1 - \delta)$; Buck-Boost $V_{out} = -\frac{\delta}{1-\delta} V_{in}$. They are independent of load (ideal load regulation) but strongly dependent on V_{in} (poor line regulation — the controller fixes this by trimming δ). The CCM/DCM boundary sets a minimum inductance, e.g. for the Buck L_{min} follows from requiring the triangular ripple $\Delta I_L = \frac{(V_{in}-V_{out})\delta T_{SW}}{L}$ to stay below $2I_{out}$.

Appendix D

Reference data

Fixed 78xx regulators: LM7805 (5 V), LM7809 (9 V), LM7812 (12 V); LM7905 for negative outputs. Typical $V_{DO,min} \approx 2$ V, ripple rejection ≈ 68 dB, output resistance 15 m Ω to 30 m Ω , short-circuit current ~ 200 mA. **Mains:** 230 V_{RMS} $\Rightarrow V_p = \sqrt{2} \cdot 230 = 325$ V.

Appendix E

Example coverage map

Example(s)	Protocol / method	Where
Ripple rejection 68 dB; $R_{out} = 20\text{ m}\Omega$; load/line table	Read the spec (P 1.1)	Ch. 1
Transformer ratios; secondary/diode sizing	Transformer + rectifier (P 2.1)	Ch. 2
Filter $C = 2.2\text{ mF}$ (and 4.7 mF)	Ripple/ t_c/I_{peak} (P 2.2)	Ch. 2
PF Ex1 ($PF = 0.6$), Ex3 (PFC $\rightarrow 1$), Ex2 (pure L , $C = 663.8\text{ }\mu\text{F}$)	Power factor + PFC (P 3.1)	Ch. 3
Linear-reg efficiency table; 78xx dissipation / heat sink	Output/ η /heat (P 4.1)	Ch. 4
Buck/Boost/Buck-Boost, $V_{in} = 9 \pm 1\text{ V}$	Topology + δ (P 5.1)	Ch. 5
Min L for CCM (Buck); Boost design; Fly-back; Boost-PFC	CCM check (P 5.1) / variants	Ch. 5
Bench: CCM/DCM test, load regulation, ripple, efficiency sweep	Characterise buck (P 6.1)	Ch. 6